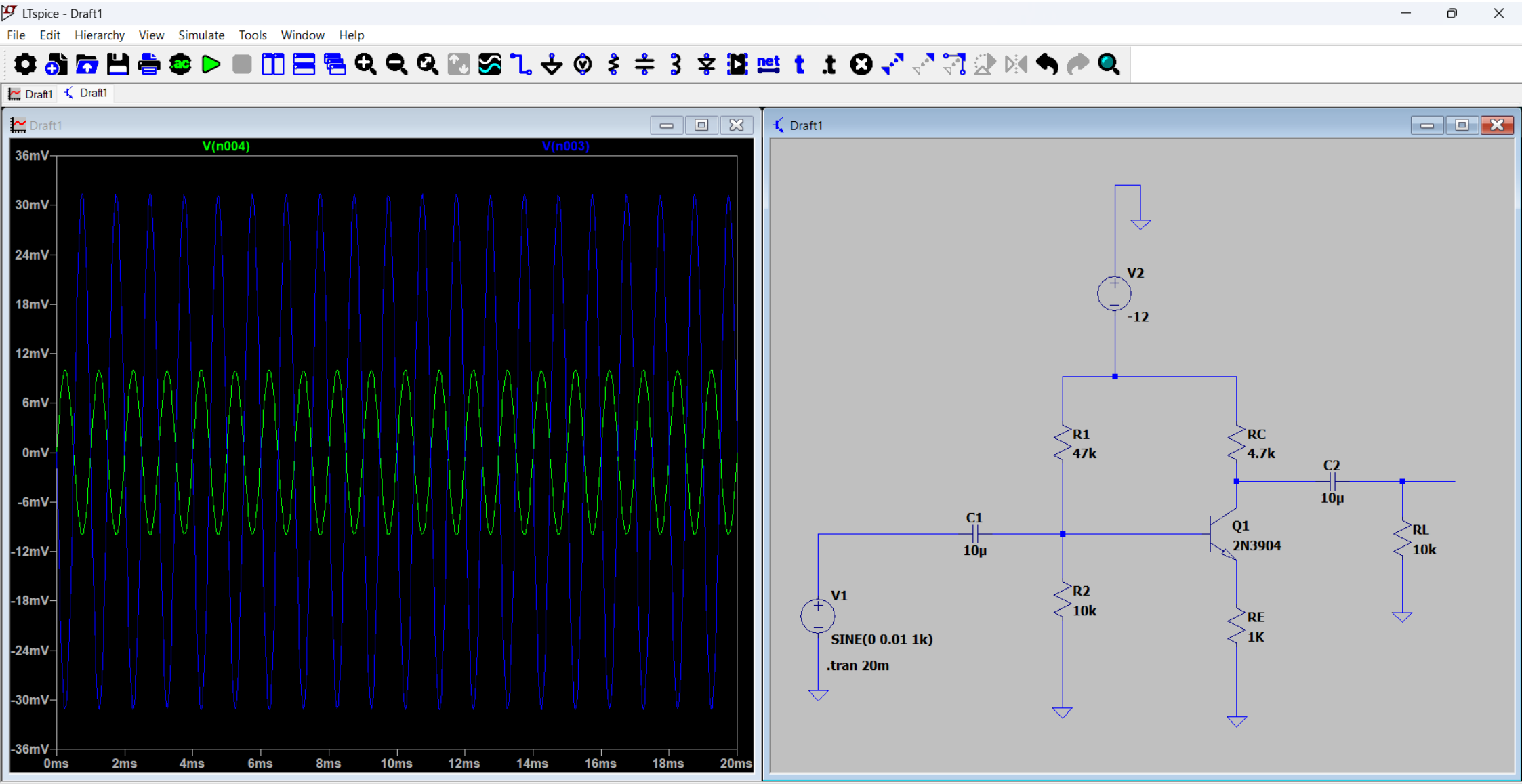
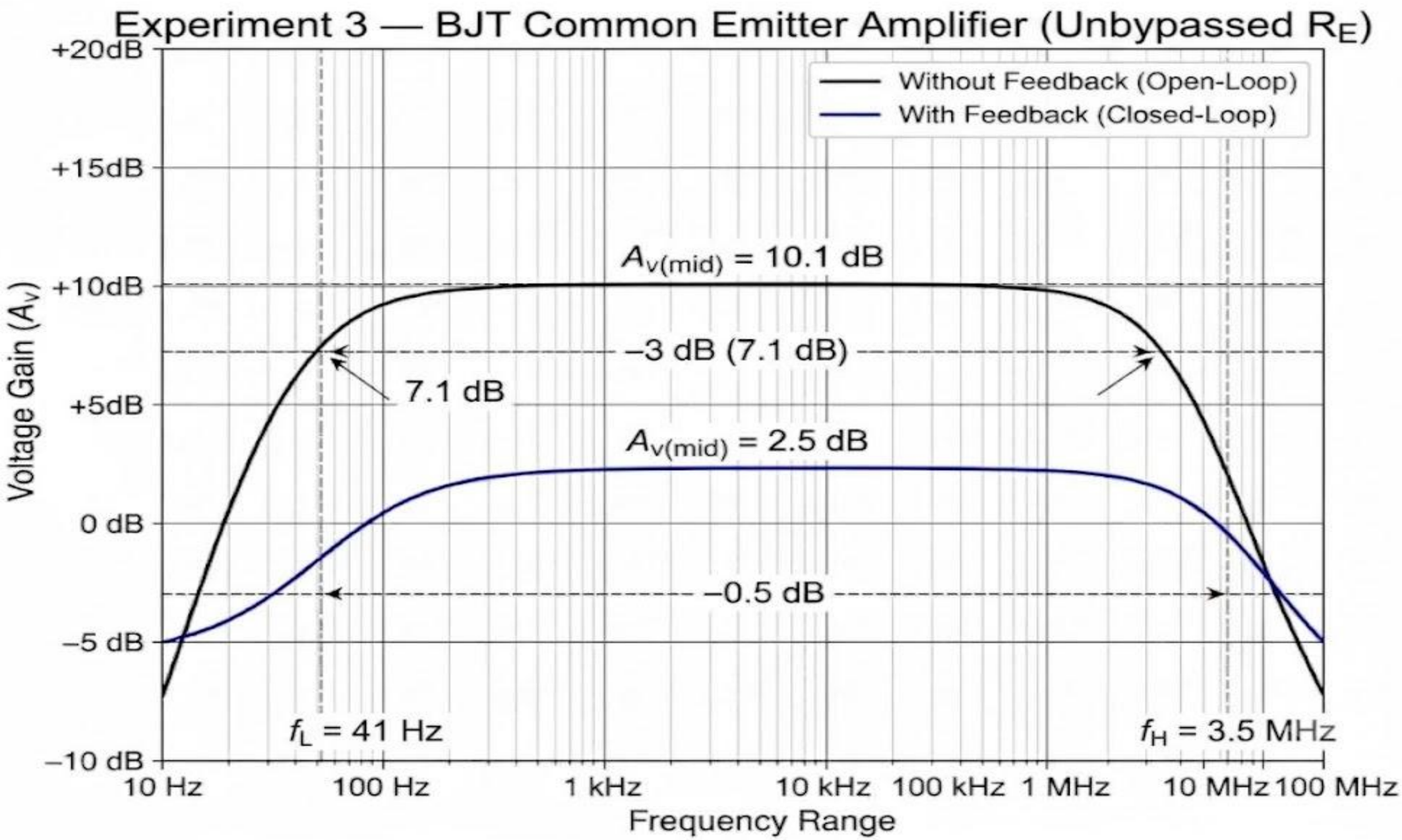

LTspice - Draft1
File Edit Hierarchy View Simulate Tools Window Help
Draft1
Draft1
Draft1
V(n004)
V(n003)
36mV
30mV
24mV
18mV
12mV
6mV
0mV
-6mV
-12mV
-18mV
-24mV
-30mV
-36mV
0ms 2ms 4ms 6ms 8ms 10ms 12ms 14ms 16ms 18ms 20ms
Draft1
V2
-12
R1
47k
RC
4.7k
C2
10µ
C1
10µ
Q1
2N3904
RL
10k
R2
10k
RE
1K
V1
SINE(0 0.01 1k)
.tran 20m


Experiment 3 — BJT Common Emitter Amplifier (Unbypassed R_E)
Without Feedback (Open-Loop)
With Feedback (Closed-Loop)
+20dB
+15dB
+10dB
+5dB
0 dB
−5 dB
−10 dB
Voltage Gain (A_v)
$A_{v(mid)}$ = 10.1 dB
−3 dB (7.1 dB)
7.1 dB
$A_{v(mid)}$ = 2.5 dB
−0.5 dB
f_L = 41 Hz
f_H = 3.5 MHz
10 Hz
100 Hz
1 kHz
10 kHz
100 kHz
1 MHz
10 MHz
100 MHz
Frequency Range